PROJECT REPORT

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1. Introduction

This project creates and assesses two machine learning systems based on data from the U.S. Census Bureau:

a) A binary classification model to determine whether income levels exceed or fall short of $50,000.
b) An analysis of customer segmentation to pinpoint unique population segments for tailored marketing strategies.

2. Dataset Overview

The analysis utilizes the dataset containing 199,523 records with 42 features spanning demographics, employment, education, financial attributes etc.

Target Variable: Binary income classification (- 50000. / 50000+.)
Class Distribution: Highly imbalanced with approximately 94% earning below $50,000

3. Data Exploration and Preprocessing

Data Quality:

- **Missing Values**: In addition to “Hispanic origin” column having Nan Values , Several features columns contained '?' symbols representing missing data, particularly in migration related columns and country of birth fields (father, mother, self)

  <u>Handled By:</u>

  - Replaced “?” with Nan in columns such as [‘migration code-change in msa’, ‘migration code-change in reg’, ‘migration prev res in sunbelt’, ‘country of birth father’].
  - Imputed Mode of column in Hispanic Origin column
- **Class Imbalance**: There was severe imbalance with 93.8% negative class (< $50,000) and 6.2% positive class (≥ $50,000)

  <u>Handled By:</u>
  - **Scale Pos Weight** - Automatically rebalanced the loss function by applying a 15:1 weight ratio to penalize minority class (income >50K) misclassifications more heavily, ensuring the model learned to identify high earners rather than defaulting to predicting the majority class
  - **Sample weights** - Incorporated census survey weights during training to maintain statistical representativeness of the population while simultaneously addressing class imbalance, creating a model that respects both individual sample importance and fair treatment across income groups

- **High Cardinality**: Many features such as age,industry codes,wage per hour, Capital gains/losses contained hundreds of unique values.

  <u>Handled By:</u>
  - Grouping age values into buckets
  - Dropping unnecessary high cardinality columns from training

- **Feature Redundancy**: Multiple highly correlated features such as detailed industry code, Num persons worked for employer etc. were removed from training as part of feature selection.

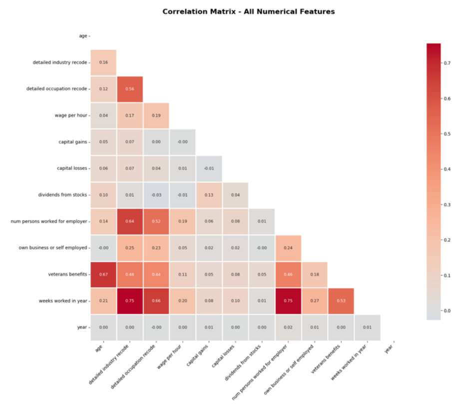
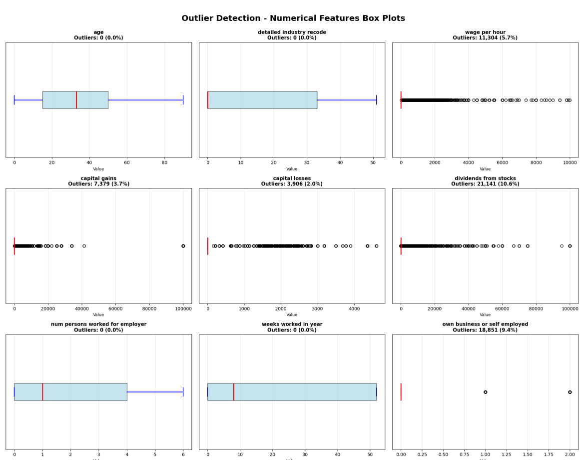


Image 1 : Numerical Features Correlation Map

- **"Not in Universe" option** - many columns had most of the data as Not in Universe(Not applicable) , such columns were dropped based on domain knowledge and correlation analysis

Visualization analysis or graph analysis:



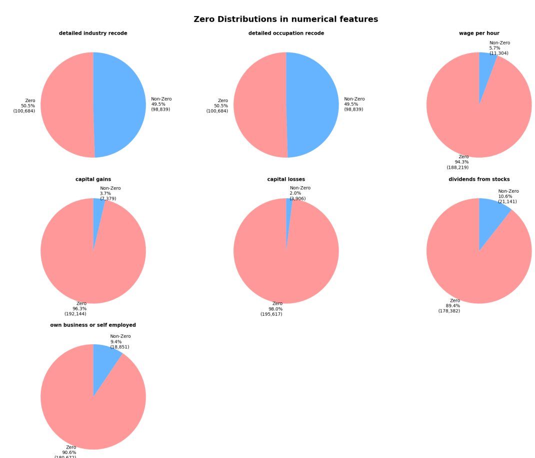
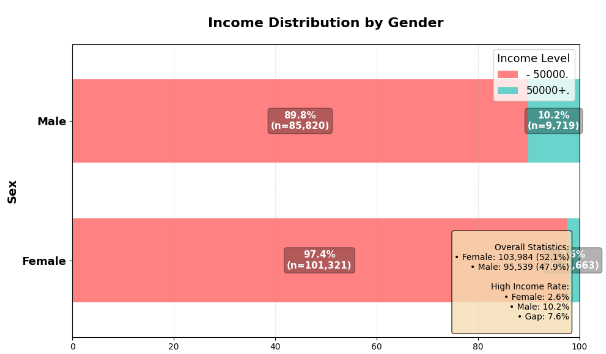


Image 2 : Outlier Detection for Numerical Features **Image 3 : % Zero's in Numerical Features**



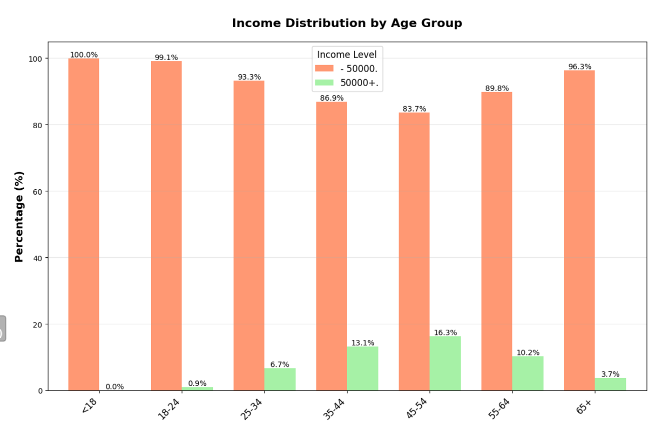


Image 4 : Income Distribution by Gender **Image 5: Income Distribution by Age**

Key Business Observations:

1. **Population Composition:**
   a. 50% are non-working individuals - The Dataset includes children, retirees, students, and unemployed individuals
   b. 45% are standard W-2 employees - Salaried workers without significant investment portfolios
   c. Only 5% are high-income/entrepreneurial individuals
2. **Employment Patters:**
   a. 94% earn salaried income - Most workers receive fixed monthly compensation rather than hourly pay
   b. 91% work as employees as against 9% who are self-employed - Traditional employment dominates, entrepreneurship is rare
   c. Mean weeks worked is 24 - This shows that many of them are either unemployed (0 weeks) or work full-time (52 weeks)
3. **Investment Patterns:**
   a. Only 11% own dividend-paying investments - Vast majority don't profit from stock/property sales
   b. Dividends from stock has large outliers signifying that there are few wealthy individuals with significantly large investment income.
4. **Demographics:**
   a. 10.2% of males earn >$50K as against only 2.6% of females
   b. Peak earning ages is 35-54 years

4. Feature Engineering

The below features were engineered to enhance the model's performance and create meaningful insights:

- **age_group:** Categorical age ranges (<18, 18-24, 25-34, 35-44, 45-54, 55-64, 65+)
- **net_capital**: (capital gains - capital losses) which represents net investment income details
- **education_level**: Ordinal encoding of education (0=Children through 16=Doctorate) to maintain the educational hierarchy
- **wage_per_hour_capped**: Restricted extreme wage values to minimize the influence of outliers
- **dividends_capped**: Capped dividend earnings to the 99th percentile
- **is_self_employed**: Binary indicator derived from "own business or self employed" field
- **has_veteran_benefits**: Binary indicator for veteran benefit recipients
- **country_of_birth_region**: Combined country of birth into larger geographic regions (United States, Latin America, Europe, Asia, Other) for father, mother, and self
- **has_migration_data**: A flag indicating whether migration information was present

Note: These transformations reduce dimensionality, handle outliers, and create interpretable features aligned with business understanding. For instance, net capital provides a single metric for investment income rather than separate gains and losses.

5. Feature Selection

Classification Task: I retained 22 features for training after removing:

- Redundant features (raw age, education, individual capital components)
- High-cardinality features (detailed occupation/industry codes, state)
- Target-related features (label, weight)
- Features with excessive missing data or low variance

Segmentation Task: I further refined to 14 features by removing:

- Region of previous residence and major industry code (high cardinality)
- Granular country of birth fields (retained only regional aggregations)
- Correlation Analysis: Identified strong relationships between:
  - Education level and veteran benefits (r = 0.84)
  - Weeks worked and number of persons worked for employer (r = 0.75)
- Since 90% of individuals had all three country of birth fields from the same region, I consolidated to a single regional indicator to reduce multicollinearity.

6. Data Preprocessing Pipeline

- **Treatment of Missing Values** : Replaced "?" with 'Unknown' if categorical variables and used median imputation for numerical features
- **Encoding**: Applied Label Encoding on categorical variables (class of worker, marital status, race, sex, etc.) to convert them into numerical format for model training
- **Scaling**: Standardized numerical features using StandardScaler to ensure they have zero mean and unit variance
- **Train-Validation-Test Split**: 60-20-20 split maintaining class distribution

7. Model Development

7.1 Classification Model Architecture

XGBoost Architecture: Tree-based ensemble model with sequential boosting

- n_estimators: 200
- max_depth: 6
- learning_rate: 0.1
- scale_pos_weight: 15.12
- subsample: 0.8
- colsample_bytree: 0.8
- random_state: 42

Cross-Validation: Implemented 5-fold stratified cross-validation to ensure robust model performance estimates across different data splits.

Model Evaluation

Model	Precision	Recall	F1-Score	Accuracy
Performance with default Threshold (0.5)				
<50k	99%	86%	92%	87%
>=50k	30%	89%	45%	87%
Performance with optimal threshold (0.85)				
<50k	97%	96%	97%	95%
>=50k	60%	60%	60%	95%

Table 1: XGBosst Classification Model Performance Summary

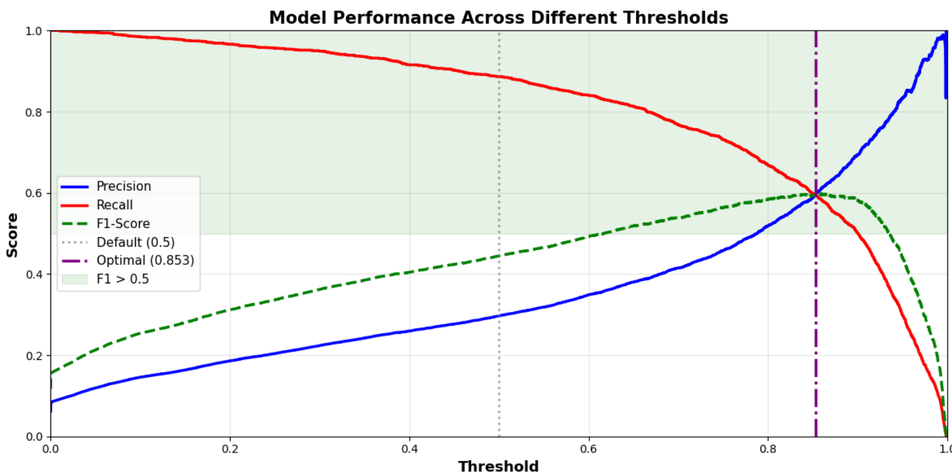


Image 6: XGBoost Model Performance

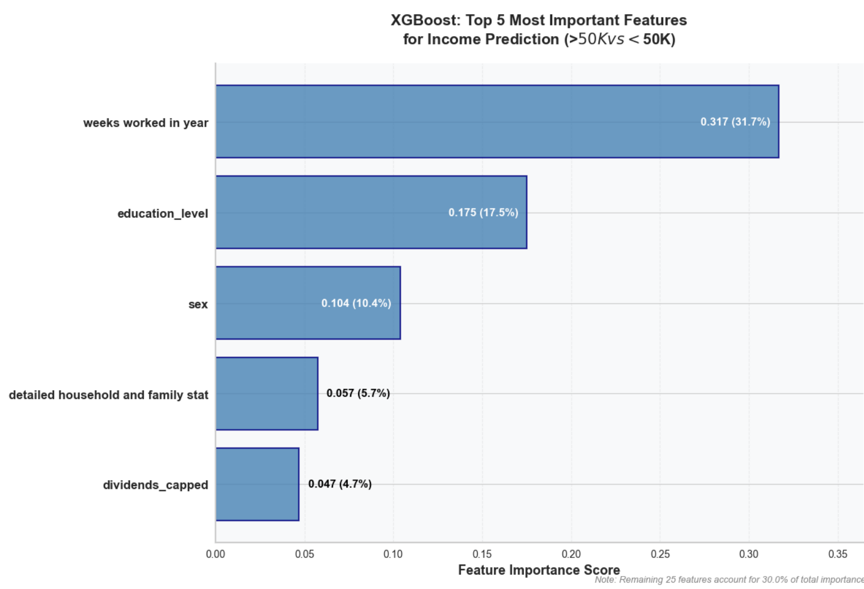


Image 7: Feature Importance

7.2 Customer Segmentation Analysis

7.2.1 Clustering Methodology

Algorithm: K-Means clustering with Euclidean distance

Reasoning: K-Means is efficient in computation for extensive datasets and generates interpretable clusters based on centroids, making it ideal for segmentation problems.

7.2.2 Optimal Cluster Determination

Assessed cluster quality through the given three metrics:

K	Silhouette Score	Calinski-Harabasz	Davies-Bouldin
2	0.19	35,368	1.59
3	0.21	29,526	1.95
4	**0.22**	**24,802**	**1.95**
5	0.22	20,253	2.15
6	0.24	18,810	1.8

Table 2: Optimal K Value Comparison

Optimal Choice: K = 4 clusters offers a solid baseline that strikes a balance between statistical integrity and business clarity, presenting well-defined segments that correspond to specific socioeconomic groups while ensuring practical application for demographic examination and marketing tactics

7.2.3 Dimensionality Reduction Visualization

Applied PCA (Principal Component Analysis) for dimensionality reduction and visualization:

- The first two principal components captured 26% of total variance (PC1: 18.1%, PC2: 8.2%), representing core socioeconomic dimensions.
- PC1 primarily captures employment status, income level, and work intensity, while PC2 reflects age, education level, and life stage characteristics

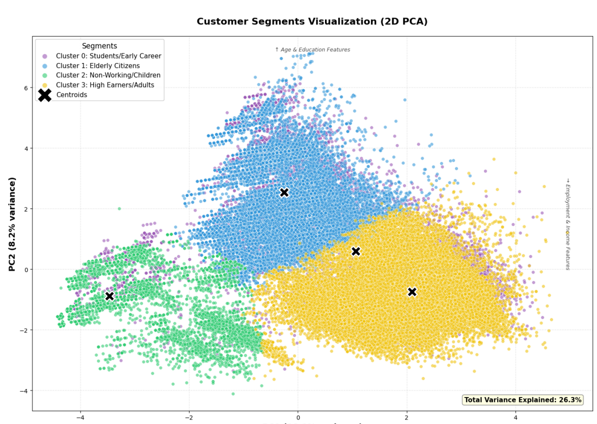


Image 8: PCA 2D Visualization

7.2.4 Segment Profiles

Cluster	Segment Name	Size	Age Group	Avg Wage/hr	Key Characteristics
0	**Students & Entry-Level**	7.6%	25-34	$51.88	- Avg hourly wage: $52 - 45% private sector workers - Net capital: $500 - High immigrant representation (37% Latin America, 20% Asia)
1	**Senior Citizens**	20.7%	65+	$0.69	- Avg hourly wage: $0.69 - 54.9% married - Net capital: $191 - 97% not in labor force - 3% receiving veteran benefits
2	**Children/Minors**	27.5%	<18	$2.57	- Avg hourly wage: $2.57 - 99% never married - 89% living as children in households - Minimal economic activity
3	**Adults with Families**	44.1%	35-44	$112.19	- Highest earners: $112.19/hr - Net capital: $725 - 71% private sector - 60.7% married with families - Most educated (avg level: Bachelor's Degree)

Table 3: Segment Profile Analysis

8. Interesting Findings and Insights

a. **Geographic Heritage Patterns**- Most families share the same birth region. 90% of individuals have parents and themselves all born in the same region, while only 5% have mixed-region parents, and less than 1% have all three from different regions.

b. **Education as an Income Predictor -** Education level emerged as the dominant feature (18% importance), with doctorate holders showing 8.5x higher income likelihood than high school graduates. This validates human capital theory and suggests educational attainment remains the strongest socioeconomic mobility driver.

c. **Gender and Marital Status Interactions**
   i) Analysis revealed that :
      (1) Married men show 3.2x higher income rates than never-married men
      (2) This effect is stronger for men than women
      (3) Divorced individuals have intermediate income levels

      Hypothesis: Family composition, conventional gender roles, and dual-income impacts earning patterns in addition to personal traits.

d. **Self-Employment Income Variance**
   i) Higher income variability (std dev 2.3x larger)
   ii) Age correlation: self-employment success increases with experience

Business Insight: Self-employed individuals require different or additional risk assessment models accounting for income volatility.

9. Business Recommendation - Marketing Strategy

Strategy	Segment 0 (Students/Early Career)	Segment 1 (Senior Citizens)	Segment 2 (Children/Minors)	Segment 3 (Adults with families)
Product Focus	▪ Entry-level credit cards ▪ Career development tools ▪ Student loans	▪ Healthcare products ▪ Retirement planning ▪ Leisure travel packages	▪ Educational toys ▪ Family entertainment ▪ School supplies	▪ Premium credit cards ▪ Investment portfolios ▪ Real estate ▪ Family vacations
Pricing	▪ Competitive pricing ▪ Payment plans ▪ Introductory discounts ▪ Bundle deals	▪ Senior discounts (15-20%) ▪ Subscription models ▪ Value-based pricing	▪ Educational discounts ▪ Bulk family rates	▪ Premium pricing acceptable ▪ Loyalty rewards ▪ VIP tiers ▪ Family packages
Marketing Channels	▪ Social media (Instagram, Facebook) ▪ Mobile apps ▪ Digital advertising ▪ Multilingual campaigns	▪ TV/Print media ▪ Direct mail ▪ In-person consultations ▪ Community centers	▪ Parental channels ▪ Schools/PTAs ▪ YouTube Kids ▪ Family websites	▪ Omnichannel ▪ LinkedIn ▪ Family events ▪ Premium magazines

Table 4 : Marketing Strategy

10.Conclusion:

The classification task achieved 95.5% accuracy using XGBoost, while the segmentation analysis identified four meaningful customer segments representing different life stages and economic profiles. These models provide actionable insights for targeted marketing, risk assessment, and policy recommendations.

11.References

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